Adversarially Robust Traffic Signal Control Under Sensor Observation Cyberattacks

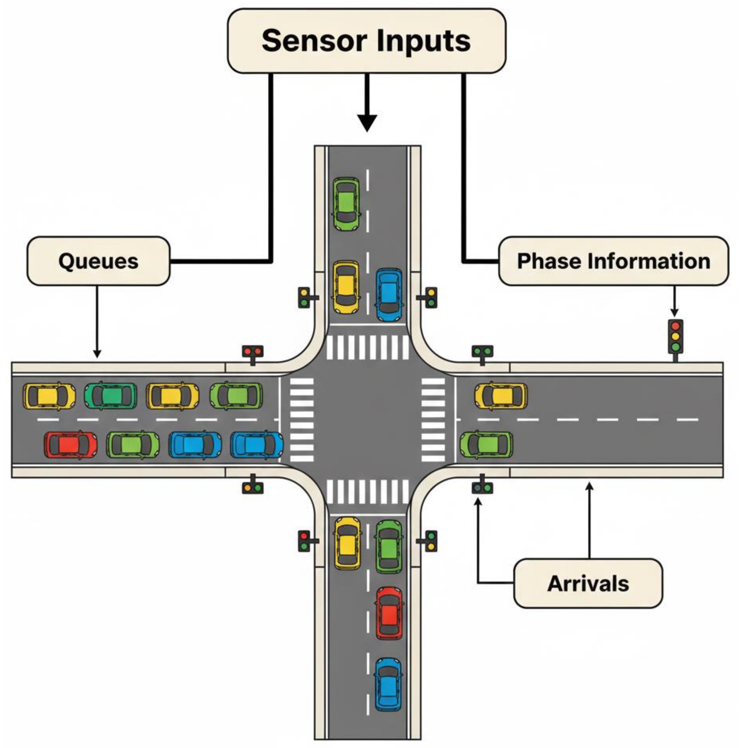
Oziel Sauceda, Mark Hernandez, Mohamadhossein Noruzoliaee, Ph.D., Fatemah Nazari Ph.D.

Safety21 INNOVATING FOR SAFETY
US DOT National University Transportation Center for Safety
NSF
The University of Texas Rio Grande Valley

Introduction

✓ Motivation

- **Adaptive** traffic signal control delivers real-world benefits over fixed-time control (e.g., 10-20% travel time and 15-30% delay reductions in an FHWA study)
- **Deep reinforcement learning (DRL)** enables efficient adaptive signal control, delivering substantial performance gains



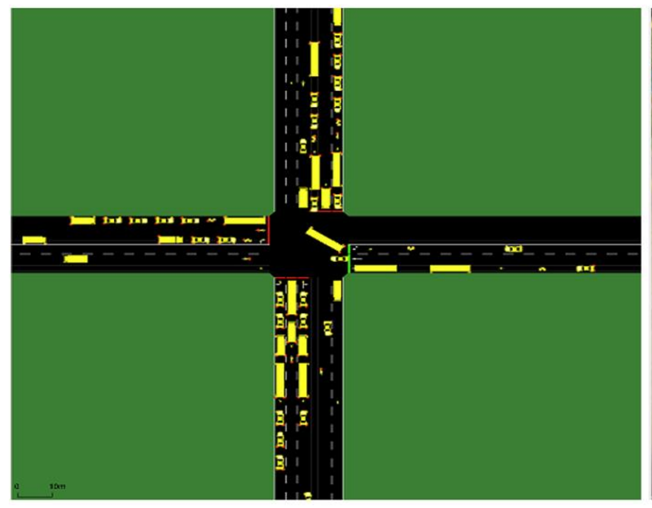
- **Cybersecurity vulnerabilities** emerge in DRL-based adaptive signal control due in part to reliance on sequential sensor observations (e.g., queues, arrivals, signal phase information)

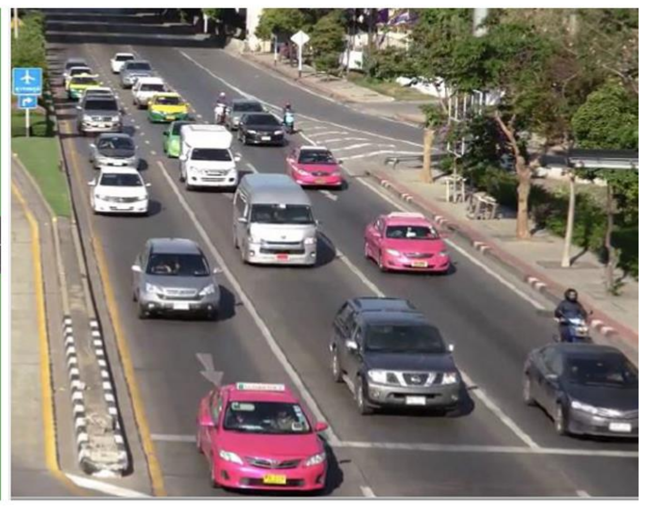
✓ Research Gap

- Due to **sequential** nature of DRL, small sensor observation perturbations can compound over time because DRL actions affect its future states
- Most DRL traffic signal research assume sensor readings are trustworthy, but there is limited work that train and evaluates **certified policies** under an adaptive (learned) attacker
- Many defenses work well on **already-seen attacks**, but a defense model that holds up even when the attacker adapts is needed

✓ Objectives and Contributions

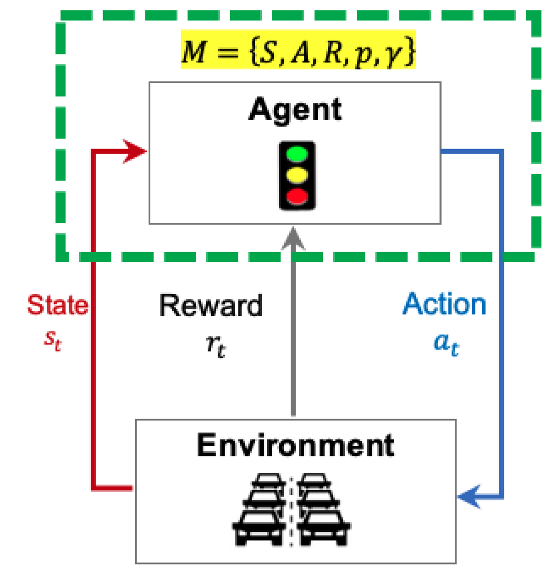
- Develop a **certified cybersecurity** defense for DRL-based traffic signal control under sensor observation attacks
- Model **attacker-defender interaction** using a State-Adversarial MDP (SA-MDP) and train using Alternating Training with Learned Adversaries (ATLA)
- Simulation experiments in SUMO on a four-way intersection

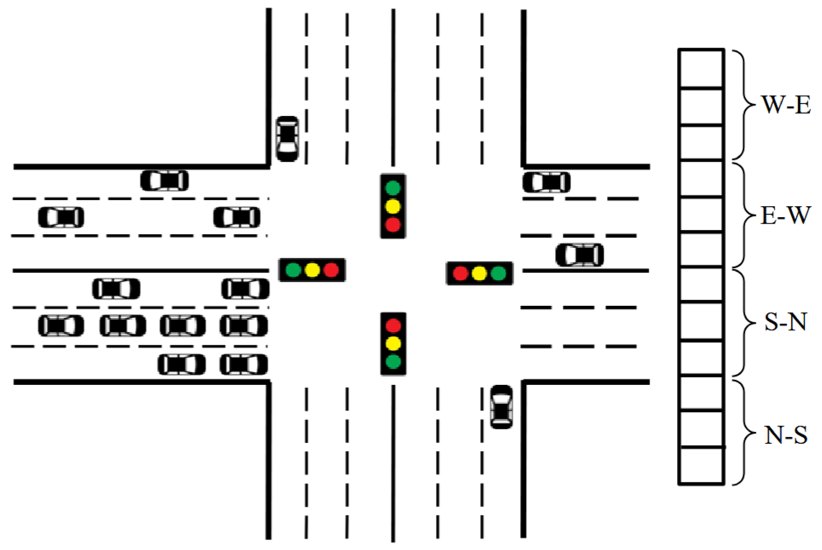




Methodology

✓ No-attack Baseline

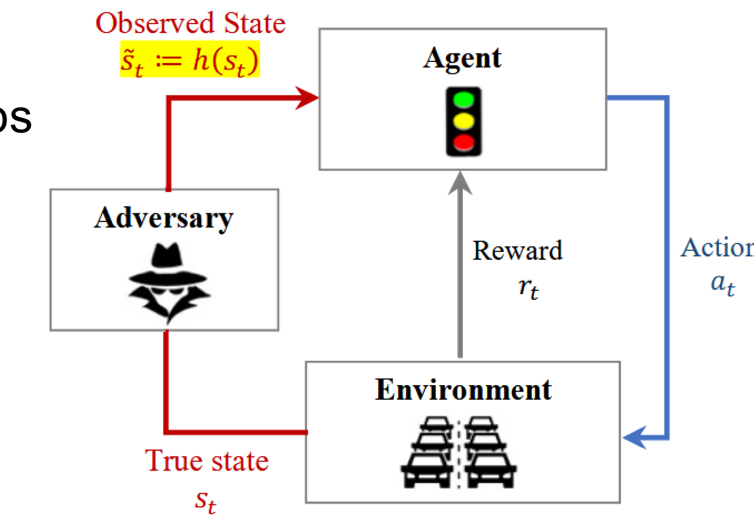




- State (S): Current phase + elapsed green + lane traffic features
- Action (A): Choose green duration each phase (fixed cyclic order)
- Reward (R): Negative intersection pressure

✓ Attack Model: Sensor Observation Attack

- No attack: DRL agent observes the true traffic state s_t
- Under attack: Attacker perturbs observations only (real-world dynamics stays unchanged)
- DRL agent observes a perturbed state: $\tilde{s} = h\,(s)$
- Attack is bounded by attack budget: $\|s - \tilde{s}\| \le \epsilon$



$$\hat{A} = S$$

$$\hat{R}(s,\hat{a},s') \coloneqq E[\hat{r}|s,\hat{a},s'] = \begin{cases} -\frac{\sum_{a\in A}\pi(a|\hat{a})\,P(s'|s,a)\,R(s,a,s')}{\sum_{a\in A}\pi(a|\hat{a})\,P(s'|s,a)} & \forall s,s' \in S;\ \hat{a}\in B(s)\subset\hat{A} \\ C & \forall s,s' \in S;\ \hat{a}\notin B(s) \end{cases}$$

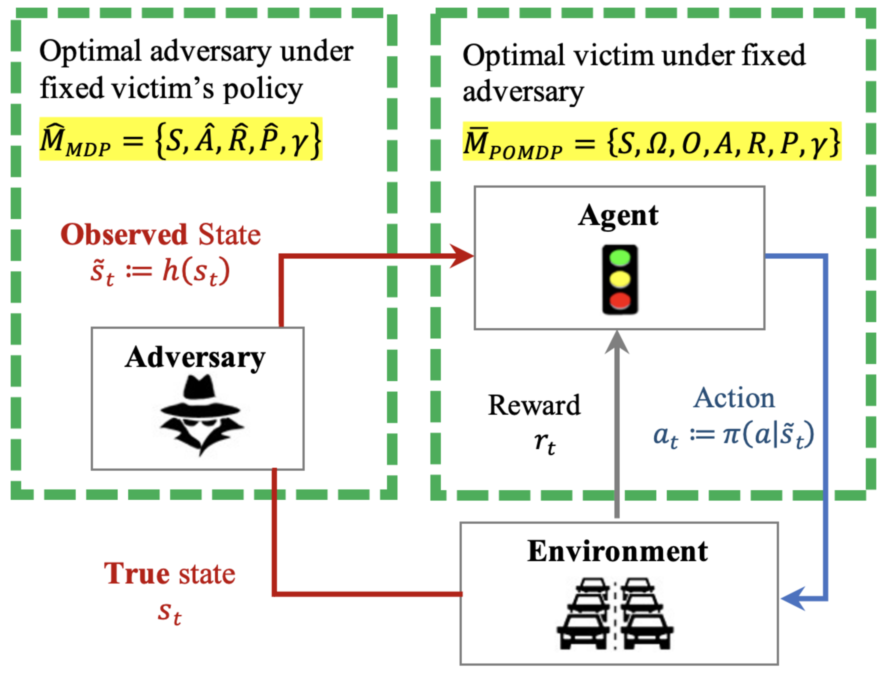
$$\hat{P}(s'|s,\hat{a}) = \sum_{a\in A}\pi(a|\hat{a})\,P(s'|s,a) \quad \forall s,s' \in S;\ \hat{a}\in\hat{A}$$

✓ Defense Model

- Adapting attack and defense through a two-player Nash game

$$O(\tilde{s}|s) = h^*(\tilde{s}|s)$$

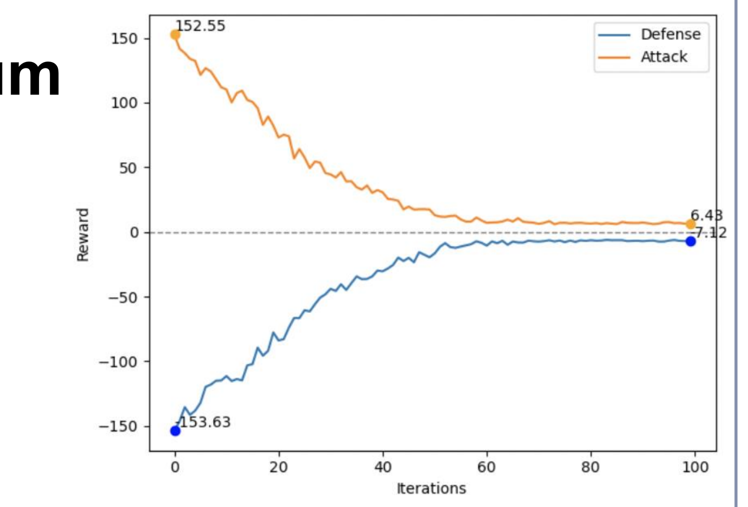
$$\Omega = \bigcup_{s\in S}\{s'|s' \in \mathrm{supp}(h(.|s))\}$$



Results

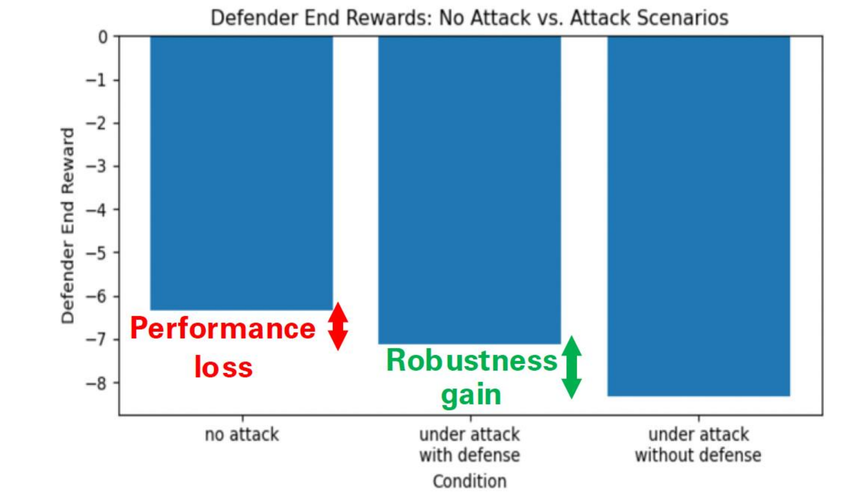
✓ Attack-Defense Equilibrium

- DRL signal agent reward: -6.35 (no-attack baseline) and -7.12 (under attack). **12%** greater intersection pressure



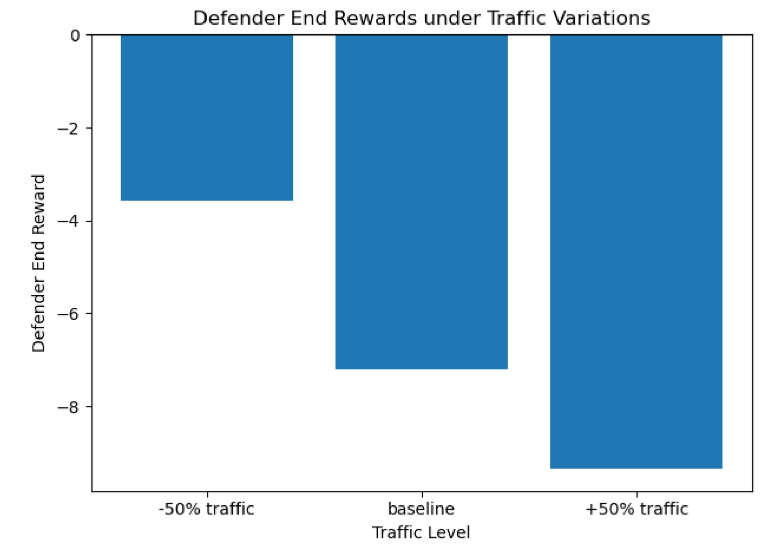
✓ Robustness-Performance Tradeoff

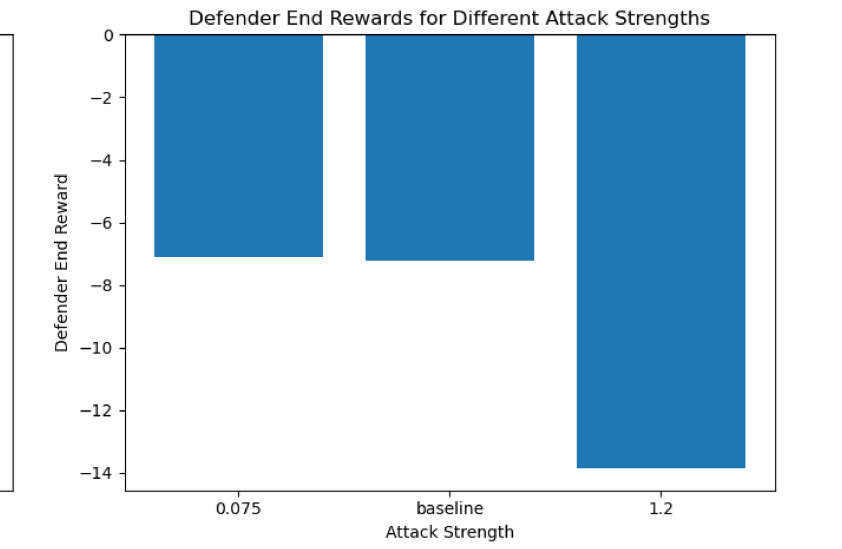
- Robustness costs **12%** nominal pressure, but prevents larger degradation (**30%**) seen in **undefended** under attack



✓ Sensitivity Analyses

- **Attack budget (ϵ):**
  - Mild ϵ = 0.075 → 2-3% change
  - Strong ϵ = 1.2 → pressure 2x but still stable
- **Traffic demand:**
  - -50% demand → -3.5 reward
  - +50% demand → -9.2 reward (25-30% more pressure), still stable





Conclusions

✓ Takeaways

- SA-MDP models optimal sensor tampering; ATLA trains a worst-case robust traffic signal controller
- Small nominal cost (~12% pressure increase), large robustness benefit vs undefended (~30% drop if undefended)
- Robust policy stays stable as attacks/demand increase (performance degrades gracefully)

✓ Future Work

- Scale to networks-level control
- Extend beyond observation attacks
- Handle unknown ϵ online estimation/adaptation